
Final Presentation Notes

Hello everyone,

we will be presenting our project, "**A Comprehensive Stock Market Database for Real-Time Analysis and Transaction Management.**"

In this project, we worked as a team to design a complete relational database for stock market data using **PostgreSQL**, tested it with real SQL queries, and created a dashboard in Tableau to visualize the results.

Why We Chose the Stock Market Domain

We selected the stock market domain because it has large amounts of data — prices, trades, news, analyst ratings that constantly change.

So, we decided to build a proper SQL database that could store, update, and analyze stock data efficiently, with the ability to support real-time queries.

Task 1: Building the Database Concept

When designing our system, we first focused on how the data

As we need to Handle millions of records from stocks, prices, users, and transactions -
Run queries like past performance analysis, sector-wise holdings, and high-value user identification

We started by building a database named **stockdb**, which includes ten tables:

- stocks
- historical_prices
- users
- transactions
- portfolios
- orders
- market_news
- financial_reports
- stock_splits

- stock_ratings

Each table was designed with careful attention to primary keys, foreign keys, and data integrity.

For example, the transactions table links users and stocks, while historical_prices records daily stock price movements.

Financial_reports capture annual company earnings for deeper analysis.

We used a combination of real and synthetic data.

- For real data, we pulled historical stock prices from Yahoo Finance for ten well-known symbols.
- For synthetic data, we generated realistic user accounts, transactions, and financial reports to ensure that each table had between three thousand and six thousand rows.

We cleaned the datasets carefully to fix missing values and ensure consistency in formats such as volume and price fields.

We made sure that all relations were normalized to Boyce-Codd Normal Form (BCNF), which helped eliminate redundancy and allowed us to maintain clean, efficient queries.

We designed our database schema starting from an E/R diagram.

We created ten key tables: - stocks - historical_prices - users - transactions - portfolios - orders - market_news - financial_reports - stock_splits - stock_ratings

Each table was linked logically using primary and foreign keys.

We enforced constraints like ON DELETE CASCADE to automatically manage related data when deleting records.

Task 3: Data Acquisition and Preparation

Task 4: Achieving BCNF (Boyce-Codd Normal Form)

One of the most important parts of our project was to make sure all relations were in BCNF.

For each table, we verified that: - Every non-key attribute is fully functionally dependent on the whole primary key - There are no partial or transitive dependencies

As a result, all our tables satisfy BCNF without any need for decomposition.

This guarantees that our database design is clean, efficient, and free from anomalies.

Task 5: SQL Query Implementation

We tested the database by writing over ten different SQL queries, covering: - Insertions: Adding new users and stocks - Deletions: Removing canceled orders - Updates: Adjusting PE ratios for utilities and increasing account balances

For selecting data, we wrote a variety of queries involving: - JOIN and ORDER BY - GROUP BY and aggregation - Subqueries - Stored functions and procedures

Examples include: - Listing users' pending orders - Summarizing sector holdings - Finding users who spent more than \$50,000 on transactions - Identifying stocks with more than five "BUY" ratings - Finding top five stocks by trading volume

Task 6: Transaction Handling with Triggers

We also implemented a safe transaction system with failure handling.

We created: - A **txn_failures** table to log failed transactions - A **trigger** that automatically raises a notification whenever a transaction fails - A **safe-buy procedure** that catches errors like negative quantities and logs them properly without corrupting the database

This demonstrates how real-world systems maintain database reliability even when something goes wrong during a transaction.

Task 7: Query Optimization

We identified three problematic queries that were initially slow: - Filtering by date and price in historical_prices - Summing transactions per user - Looking up specific holdings in portfolios

We improved performance by creating indexes: - A composite index on (date, close_price) - An index on user_id in transactions - A unique index on (user_id, stock_id) in portfolios

Using **EXPLAIN ANALYZE**, we observed that query execution times were significantly reduced after applying these indexes.

Overall Database Implementation

Through this entire process, we implemented a stock market database that:

- Supports millions of records efficiently
- Maintains strong referential integrity
- Enables both complex analytical queries and safe transactional updates - Is scalable and ready for advanced applications like real-time dashboards and predictive analytics

Dashboard Visualization with Tableau

After building the database, we connected it to Tableau for visual analysis with PostgreSQL server.

From there, we created several visualizations to explore different aspects of the stock market and user activities.

We will now walk through each visualization and explain the insights we gathered.

1. Portfolio Sector Distribution (Stacked Bar Chart)

This chart displays how each user's investment portfolio is distributed across various sectors, such as Technology, Healthcare, and Finance.

By stacking sectors within each user's bar, we can easily see how diversified or concentrated their portfolios are.

Users with a large concentration in a single sector may face higher risk during sector-specific downturns.

2. Stock Ratings Trends (Grouped Bar Chart)

In this visualization, we analyzed the number of "BUY," "HOLD," and "SELL" ratings given by financial analysts for each stock.

Grouping the ratings allows us to see overall market sentiment quickly.

Stocks with a higher number of "BUY" ratings suggest stronger positive sentiment, whereas a higher count of "SELL" ratings indicates possible concerns from the analyst community.

3. Top 5 Stocks by Market Capitalization (Horizontal Bar Chart)

This chart ranks the top five companies by their total market capitalization.

It helps highlight which firms dominate the stock market by wealth.

Companies like Apple and Microsoft usually appear at the top, demonstrating their size, stability, and influence in global markets.

4. Daily Closing Price Trends (Line Chart)

This line chart tracks the closing prices of selected stocks, including Apple, Microsoft, and Google, over time.

By plotting multiple stocks on the same timeline, we can observe how each company performs relative to the market and identify trends, rallies, or declines tied to market events.

5. User Transaction Volume Heatmap (Heatmap Table)

The heatmap visualizes user trading activity over time.

Each cell represents the amount of money a user spent on transactions for a particular day. Darker colors indicate higher volumes.

This visualization makes it easy to spot active traders and periods of high trading intensity, such as around earnings reports or major market events.

6. Financial Growth Trends (Dual Line Chart)

In this dual line chart, we compared revenue and net income trends for selected companies over multiple years.

Seeing these two financial metrics side-by-side helps us understand whether a company's profits are growing alongside its sales.

Consistent upward trends in both lines generally suggest strong business performance.

Advanced Visualizations (Optional Extensions)

To add even more depth to our dashboard, we created several advanced plots:

- A stock volatility chart showing daily price fluctuations
- A box plot of P/E ratio distributions across sectors
- A cumulative portfolio value trend chart for individual users
- A Gantt chart visualizing analyst rating changes over time
- A scatter plot comparing revenue to net income for companies

These advanced plots provide additional dimensions to financial analysis, moving beyond simple summaries into patterns of risk, valuation, and operational efficiency.

Dashboard Design Choices

We carefully structured our dashboard layout to make it clear and easy to follow:

- We grouped related charts horizontally and vertically using containers
- We kept the color scheme consistent, using blue for financial indicators and red for warning signals
- We added simple captions under each visualization to guide interpretation
- We used clean text and minimal decorations to keep the focus on the data itself

Our goal was to create a dashboard that is not only informative but also easy to explore interactively.

Publishing to Tableau Public

After building the dashboard, we published it on Tableau Public.

This allowed us to generate a shareable link that anyone can use to access and interact with the dashboard online.

By publishing publicly, we ensure that the project can be easily presented, shared, and demonstrated across different devices.

Conclusion

We believe that the methods we used here can be extended to solve even larger and more complex data problems in real-world industries.

Thank you for listening.

We are happy to take any questions.